

Crypto Code Constructions in Post-Quantum Cryptography

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Abstract — The development of a full-scale quantum computer calls into question the stability of modern cryptosystems. Thus, all asymmetric cryptosystems can be compromised (hacked) based on Shor's quantum algorithm (possibly also digital signature algorithms on asymmetric cryptosystems, including those using elliptic curve cryptography). Using Grover's quantum algorithm will be able to hack symmetric encryption algorithms, including the AES-128 cipher. In the conditions of the post-quantum period of evolution, algorithms of post-quantum cryptography are developing. Among the winners of the NIST competition for algorithms of post-quantum cryptography, a special place is occupied by crypto-code structures that not only ensure authenticity, but are also capable of integrated confidentiality and integrity. At the same time, the use of noise-resistant codes allows "additional" provision of reliability. In addition, in terms of the speed of cryptographic transformations, crypto-code structures provide a speed at the level of modern algorithms of symmetric cryptography.

Keywords — *crypto-code constructions, post-quantum cryptography, algebraic geometric codes.*

I. INTRODUCTION

The evolutionary development of computing capabilities, the synthesis of mobile and smart technologies, on the one hand, allows for the automation of processes in various areas of IT [1–3]. On the other hand, they require more stringent measures to counteract quantum hacking algorithms. In addition, the synthesis of smart technologies with artificial intelligence allows for a significant "simplification" of analysis issues and a reduction in decision-making time. However, the use of neural networks does not provide the required level of "objectivity" in the results obtained, and attacks on neural networks cast doubt on their use in intelligent information security systems.

In recent decades, computing power has grown rapidly, in line with the predictions of Gordon Moore (founder of Fairchild Semiconductor and Intel) based on his empirical observations. According to Moore, computer system performance doubles every 18 months, resulting in a 3-5-fold increase (i.e., 1,000-100,000 times) over a period of 5-10 years. If this dynamic continues, computing devices will show exponential growth in power in a relatively short time.

This leap in computing power provides attackers and cybercriminals with the resources to attack the cryptographic algorithms on which the security of services depends. In addition, in 1994, American mathematician Peter Shor presented an algorithm that allows factoring an integer of arbitrary length into prime factors almost as fast as multiplication. Shor's algorithm factorizes a number N in $O(\lg^3 N)$ time using an $O(\lg N)$ -bit register, significantly outperforming any classical factorization methods [49].

One of the main advantages of quantum registers is significant memory savings: N quantum bits are capable of storing an amount of information equivalent to $2N$ bits, and the interaction between qubits allows for a single operation to affect the entire register (the principle of quantum parallelism). Due to this, Shor's algorithm has cast doubt on the reliability of asymmetric cryptographic systems, since it can be used to effectively solve the discrete logarithm problem and other problems underlying these algorithms.

These findings were confirmed in March 2018 in a report by experts from the National Institute of Standards and Technology (NIST) of the United States (Report on Post-Quantum Cryptography) [1, 3–7]. In 2019, NIST experts, who announced a competition for post-quantum cryptographic algorithms, also pointed out the vulnerability of algorithms based on elliptic curves [1, 3–7]. Thus, humanity is entering a post-quantum period, when traditional encryption methods will be significantly improved, and quantum computers with the necessary register sizes (in qubits) and the corresponding mathematical and software will become a reality for effective cryptanalysis.

Key problems that can be solved on quantum computers include [1, 3–7]:

- Shor's factorization algorithm;
- Grover's algorithm for finding an element in an unordered database;
- Shor's algorithm for solving the discrete logarithm problem in a finite field;
- Shor's algorithm for finding the discrete logarithm of a group of points on an elliptic curve;

- Quantum cryptanalysis algorithms aimed at transforming a problem into a ring factor;
- The quantum cryptanalysis algorithm proposed by Xiong and Wang, with its further improvements.

Table 1 contains a comparative analysis of the factorization complexity for classical and quantum algorithms [1, 3–7], and Table 2 demonstrates the complexity of implementing Shor's method for taking discrete logarithms in a group of points on an elliptic curve [1, 3–7].

TABLE I. FACTORIZATION COMPLEXITY ANALYSIS FOR CLASSICAL AND QUANTUM ALGORITHMS

Module size N, bit	Number of qubits $2n$	The complexity of the quantum algorithm $4n^3$	The complexity of the classical algorithm
512	1024	$0.54 \cdot 10^9$	$1.6 \cdot 10^{19}$
3072	6144	$12 \cdot 10^{10}$	$5 \cdot 10^{41}$
15360	30720	$1.5 \cdot 10^{13}$	$9.2 \cdot 10^{80}$

TABLE II. THE COMPLEXITY OF IMPLEMENTING THE SHOR METHOD OF DISCRETE LOGARITHMIZATION FOR A GROUP OF POINTS OF AN ELLIPTIC CURVE

Algorithm for developing a discrete logarithmic equation			
Base point order size, bits	Number of qubits required $f(n) = 7n + 4 \log_2 n + 10$	Complexity of quantum algorithm $360n^3$	Complexity of the classical algorithm
163	1210	$1.6 \cdot 10^9$	$3.4 \cdot 10^{24}$
256	1834	$6 \cdot 10^9$	$3.4 \cdot 10^{38}$
571	4016	$6.7 \cdot 10^{10}$	$8.8 \cdot 10^{85}$
1024	7218	$3.8 \cdot 10^{11}$	$1.3 \cdot 10^{154}$

The results of the comparisons presented in Tables 1 and 2 indicate a significant reduction in energy costs for the implementation of hacking cryptographic algorithms of asymmetric cryptography, which include algorithms for electronic digital signatures.

To ensure security in the post-quantum period, NIST specialists in the USA propose using post-quantum algorithms that are capable of providing 5 levels of security. In addition, the analysis of the winners of the competition for post-quantum algorithms shows the need to synthesize the theory of noise-resistant coding based on the mathematical apparatus of Galois rings (fields), with the theory of cryptographic transformations. Such a synthesis provides the ability to resist the quantum algorithms of Grover and Shor. Among the recommended asymmetric cryptosystems of post-quantum cryptography, a special place is occupied by the crypto-code constructions (CCC) of McEliece on Goppa codes, which allow simultaneously providing the required level of security, the speed of transformations (coding) at the level of the speed of symmetric algorithms and reliability due to the use of noise-resistant codes as key data.

II. MAIN RESEARCH

Cryptographic code structures (CCC) provide protection of information from accidental and deliberate attacks, which was first proposed in studies [9–11]. The basic principle of such structures is based on the use of an algebraic block (n, k, d) code, which has efficient encoding and decoding algorithms. By disguising an algebraic code as a random code (general state code), it is possible to transform the task of

decoding it into a computationally difficult problem for an attacker. Indeed, without knowledge of the masking rule, the attacker will have to use a complex random code decoder, which makes the encoding-decoding process equivalent to a one-way cryptographic function [11–13]. Thus, the basis of CCC is the difficulty of decoding a random code from a complexity-theoretical point of view.

The choice of an algebraic (n, k, d) -block code is determined by the complexity of the coding and decoding algorithms. Preference is usually given to codes whose complexity of these algorithms depends on the length of the code and/or its correcting ability in a polynomial manner. Among such codes, one can distinguish BCH codes, PC codes, Goppa codes, algebraic geometric codes, and others.

The process of masking an algebraic code consists of presenting it to an attacker as a random code (a common state code). In general, the complexity of decoding a random code increases exponentially with its length and/or error-correcting ability. The asymmetric crypto-code constructions of McEliece and Niederreiter use masking matrices as a private key, and the public key is the generating matrix in McEliece's CCC, and the parity-check matrix of the error-correcting code in Niederreiter's CCC. In addition, Rao-Nam's CCC forms a symmetric system in which the secret key is the generating matrix (in the modified Rao-Nam CCC, a permutation matrix is additionally used) [10]. However, with all positive parameters of the CCC, a method for hacking the private key is proposed in [13] - finding the elements of the generating matrix. In addition, options for eliminating this attack are proposed - using cascade or algebraic geometric codes. Due to the orthogonality of the generating and checking matrices, this attack can be implemented on all CCCs.

To implement the CCC on algebraic geometric codes, the works [18–20] propose using algebraic curves of the 3rd kind (elliptic curves), which make it possible to obtain elliptic codes with better code characteristics when transmitting more than 1000 bits than Reed-Solomon codes.

An algebraic geometric code on a curve X over $GF(q)$ is a linear code of length $n \leq N$ whose code words $C(c_1, c_2, \dots, c_n)$ are given by:

$$\sum_{i=0}^{k-1} i_j F_j(P_i) = c_i$$

where $P_i(X_i, Y_i, Z_i)$ are the projective points of the curve X , i.e. (X_i, Y_i, Z_i) are the solutions of the homogeneous algebraic equation defining the curve X , $i = \overline{1, n}$, $F_j(P_i)$ are the values of the generator functions at the points of the curve. In this case, an elliptic curve (EC) in the affine space A^2 over the field $GF(q)$ is a smooth curve defined by the equation:

$$y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6,$$

or in P^2 given by a homogeneous equation

$$y^2 z + a_1 xyz + a_3 yz^2 = x^3 + a_2 x^2 z + a_4 xz + a_6 z^3, \\ a_i \in GF(q), \text{ curve genus } g = 1.$$

Let us consider the McEliece crypto-code construction, the structural diagram is presented in Fig. 1.

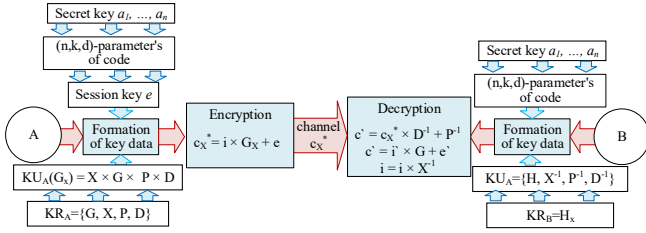


Fig. 1. Protocol for information exchange using McEliece crypto-code construction on the EU

Let G be a generator matrix of a linear (n, k, d) code over $GF(q)$ with polynomial decoding complexity. Let X be a non-singular $k \times k$ -matrix over $GF(q)$, D be a diagonal matrix with non-zero elements on the diagonal, and P be a permutation matrix of size $n \times n$. The permutation matrix implements a permutation of the coordinates of a vector as a matrix multiplication, namely, the element p_{ij} of the matrix P is equal to 1 if and only if the coordinate with the number i is mapped by means of a permutation to the coordinate with the number j . In other cases, $p_{ij} = 0$. Thus, the matrix P contains only one unit in each column and in each row. The product of the matrices $A = P \cdot D$ defines the permutation matrix A with non-zero elements of the field $GF(q)$. The permutation matrix A (unipotent matrix) when permuting the coordinates of a vector, it preserves the Hamming distance, i.e. $d(a, b) = d(a \cdot A, b \cdot A)$, where $d(x, y)$ is the Hamming distance between vectors x and y .

The main difference in Niederreiter's CCC is the use of a verification matrix as a public key, Fig. 2 shows the CCC on the EC.

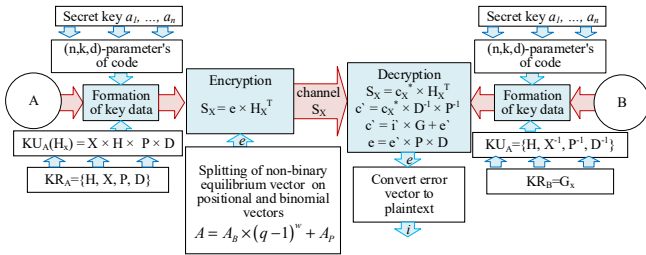


Fig. 2. Information exchange protocol using the Niederreiter crypto-code construction on the EU

Let H be a parity-check matrix of a linear (n, k, d) code over $GF(q)$ with polynomial decoding complexity. Let X be a non-singular $r \times r$ -matrix over $GF(q)$, D be a diagonal matrix with non-zero elements on the diagonal, P be a permutation matrix of size $n \times n$. The public key in the Niederreiter scheme is the matrix $H_X = X \cdot H \cdot P \cdot D$, the secret (closed) key is the matrices X, P, D . The closed information (codogram) S_X is a vector of length $r = n - k$ and is calculated by the rule:

$$S_X = e \cdot H_X^T,$$

where vector e is a vector of length n and weight $\leq t$ that carries confidential information (information message to be closed).

The symmetric Rao-Nam CCC on the EC is shown in Fig. 3.

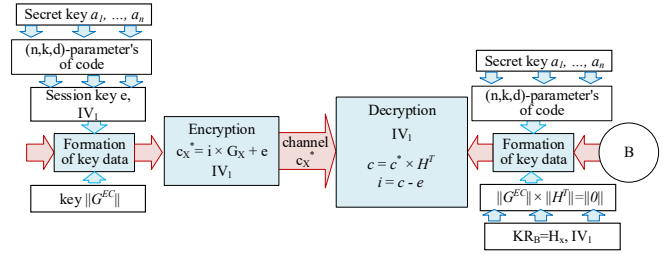


Fig. 3. Protocol for information exchange using the Rao-Nam crypto-code construction on the EU

However, the basic requirements of NIST specialists to ensure the required level of safety do not allow their use in energy-dependent technologies.

To eliminate this drawback (the complexity of implementation over the Galois field $GF(2^{10-2^{13}})$ in the works [15–17] it is proposed to use modified elliptic codes (MEC) – shortened (after forming the code word according to the initialization vector, several symbols of the code word are “removed”) and/or extended (after shortening according to the initialization vector 2, symbols of the plaintext are “added” to the shortened code word). Figures 4–5 show the McEliece CCC on shortened MEC and extended MEC.

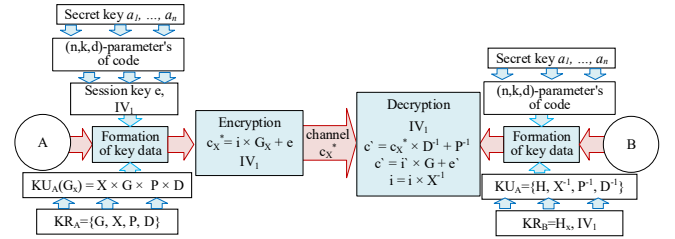


Fig. 4. Information exchange protocol using McEliece crypto-code construction on shortened MECs

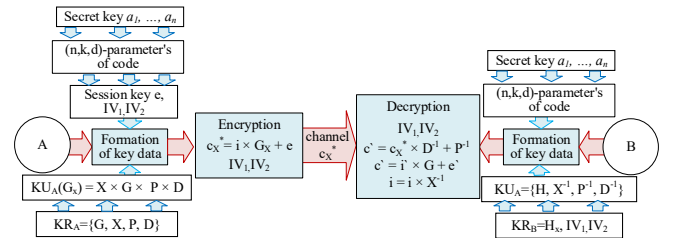


Fig. 5. Information exchange protocol using McEliece crypto-code construction on extended MECs

Figures 6–7 show the information exchange protocols using the Niederreiter CCC on the MEC.

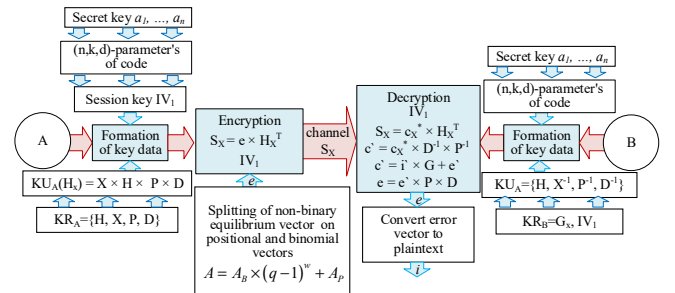


Fig. 6. Information exchange protocol using Niederreiter's crypto-code construction on shortened MECs

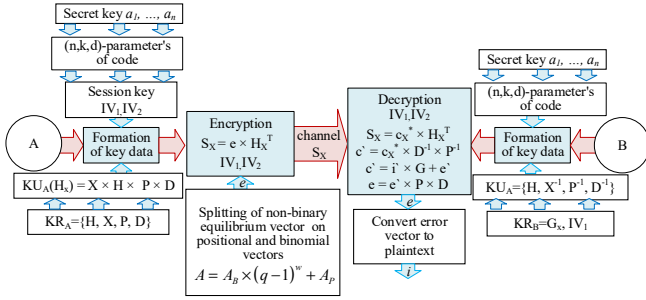


Fig. 7. Information exchange protocol using Niederreiter's crypto-code construction on extended MECs

Figures 8–9 show the protocols for exchanging information using the Rao-Nam CCC at MEC.

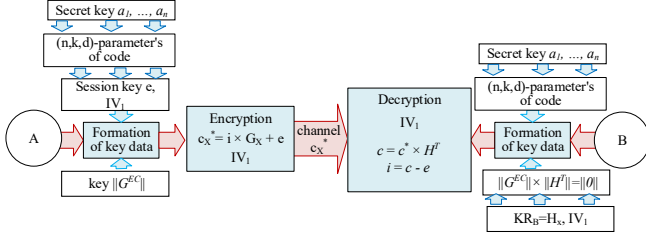


Fig. 8. Information exchange protocol using the Rao-Nam crypto-code construction on shortened MEC

The conducted stability/efficiency analysis showed that the use of CCC on MEC allows to ensure the required level of stability when constructing them over the Galois field $GF(2^6 - 2^8)$ [15–20].

The difficulty of hacking for the CCC on the EU is given by the expressions [15–17, 20]:

$$\text{— on EC: } O_{K+} = N_{\text{cover}} \times n \times r,$$

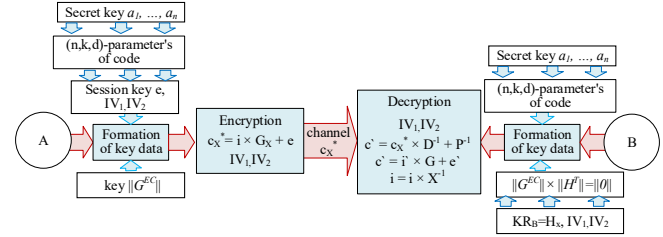


Fig. 9. Information exchange protocol using the Rao-Nam crypto-code construction on extended MECs

$$N_{\text{noxp}} \geq \frac{C_n^{\rho-t}}{C_{n-k}^{\rho-t}} = \frac{n(n-1)\dots(n-\rho-t-1)}{(n-k)(n-k-1)\dots(n-k-\rho-t-1)},$$

Where $t = \lfloor (d-1)/2 \rfloor$.

– on shortened codes:

$$O_{K+} = N_{\text{noxp}} \times (2\sqrt{q} + q + 1 - 1/2k) \times r.$$

– on extended codes:

$$O_{K+} = N_{\text{noxp}} \times (2\sqrt{q} + q + 1 - 1/2k + 1/2k) \times r.$$

Table 3 shows the results of studies of the stability of the CCC in relation to energy intensity and conversion rate.

TABLE III. BREAKDOWN COMPLEXITY RESULTS FOR THE EU (MEC)

$lg(l_s)$	Relative coding rate, R					
	0.5	0.75	0.5(el)	0.75(the)	0.5(sh)	0.75(sh)
1	4.75	12.1	15.6	18.23	19.12	19.82
2	10.52	21.76	32.47	35.67	38.63	39.18
3	18.22	33.17	43.75	51.61	56.88	58.03
4	21.42	51.75	59.43	72.81	78.92	80.52
5	38.77	61.09	68.26	87.32	94.91	104.56
6	54.13	78.37	101.72	112.46	120.83	128.79
7	82.14	83.72	156.75	164.72	182.39	189.74
8	165.84	179.13	223.64	231.57	276.27	287.33
9	358.33	371.09	421.97	428.63	459.81	476.52
10	672.37	684.94	716.41	722.26	783.46	794.28

The analysis of Table 3 showed that when using the MEC-based CCC, not only the required level of stability is maintained (initialization vectors 1 and 2), but also the computational costs are reduced, which leads to an increase in the speed of transformations by 4.5 MEC. However, smart technologies require lower-energy technologies and mechanisms.

For this purpose, hybrid CCC (HCCC) based on the use of the "ideal" compression algorithm MV2 (multi-channel cryptography on damaged codes) are proposed in [18–20]. The basis of multi-channel cryptography is the ability to separate the damage - a pseudo-random sequence that determines the number of deleted bits in each symbol of information. And the damaged text - the text formed after the damage has been inflicted. The essence of the method is to split the binary representation of the plaintext into the damaged text and the damage in order to reduce the lengths of

the text symbol codes beyond their redundancy based on the ideal compression algorithm.

Fig. 10 shows the structural diagram of MV2.

In this case, as a rule, the damage and the damaged text are transmitted through different channels, which provides an additional level of entropy for the attacker.

The difficulty of hacking for the CCC on the EU is given by the expressions [15–17, 20]:

– on shortened MEC with damage caused:

$$O_{K+} = N_{\text{noxp}} \times (2\sqrt{q} + q + 1 - 1/2k) \times r + N_{F_{\text{min}}} (N_k),$$

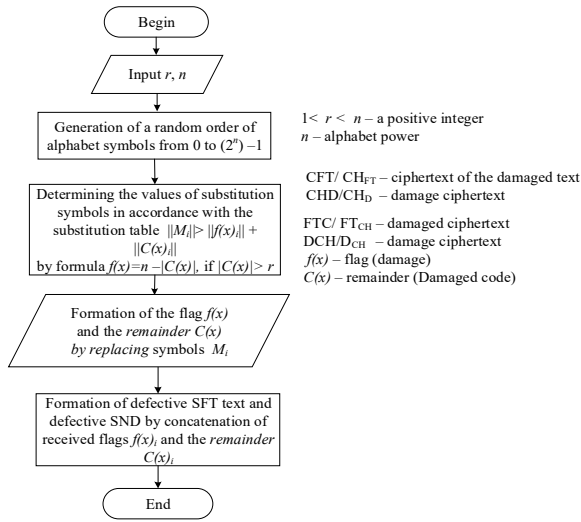


Fig. 10. Information exchange protocol using the Rao-Nam crypto-code construction on extended MECs

where $N_F \approx \frac{K_C^z}{2^{1-K_C^{z+1}}} \times |F|$; $K_C = 97/128$; $|F|$ – total length of output flags (damages) (bits) – with the remainder (loss text) known to the attacker and the given flags (damages), with an unknown key: $N_K \approx 2^{1190 \times z}$; $z = 16$;

- on shortened MEC with damage caused:

$$O_{K+} = N_{nkp} \times (2\sqrt{q+q+1-1/2k+1/2k}) \times r + N_{F_{\text{HCC}}} (N_K).$$

Table 4 presents the results of studies of the resistance of the HCCC in relation to energy consumption and resistance to burglary.

Analysis of Table 4 showed that the use of HCCC on lossy codes provides a 12-fold reduction in the number of elementary group operations when forming a cryptogram. When scattering - about 20 times the number of elementary group operations on the receiving side.

TABLE IV. HACKING DIFFICULTY RESULTS FOR MEC (MEC+DC)

Relative coding rate, R	$lg(l_s)$					
	1	2	3	4	5	6
0.5(the)	15.6	32.47	43.75	59.43	68.26	101.72
0.75(the)	18.23	35.67	51.61	72.81	87.32	112.46
0.5(sh)	19.12	38.63	56.88	78.92	94.91	120.83
0.75(sh)	19.82	39.18	58.03	80.52	104.56	128.79
0.5(elh)	7.21	9.46 p.m.	31.68	41.72	56.63	72.32
0.75(elh)	9.17	23.72	33.83	42.27	58.91	74.79
0.5(sh h)	12.54	27.48	37.38	47.48	62.86	89.5
0.75(sh h)	14.56	29.82	38.43	58.23	66.53	97.71

CONCLUSIONS

The proposed algorithms for constructing crypto-code structures by McEliece, Niederreiter and Rao-Nam allow the formation of post-quantum asymmetric and symmetric cryptosystems that provide the required level of security and ensure the implementation of all security services. At the same time, this approach allows the implementation of security services for virtually any systems and technologies, while ensuring the required level of cryptographic resistance, energy independence and the possibility of practical implementation on any chipsets.

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